Model Predictive Control Spring 2026
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Solution 02
Unconstrained Control

Exercise 1 **Finite vs Infinite Horizon LQR (Implementation)**

A1: See MATLAB code.

A2: Run the code for different values of `N` to find $\bar{N} = 7$.

A3: For short horizons, the balance between the cost of the state trajectory and the input trajectory comes down in favour of the input, since the state doesn't move "too much" over the short term if no input action is taken. Over a longer horizon, the cost of the state trajectory will dominate, causing the input to take action to decrease it.

A4: No, see example in the lecture slides (lecture 2, page 38).

B1-2: The infinite horizon LQR controller has lower cost. See MATLAB code.

Exercise 2 **Finite Horizon Optimal Control (Implementation)**

1) See MATLAB code.

2) See MATLAB code.

3) See MATLAB code. The costs are equal in the three cases, in particular:

```
costBa = 1872.3
costQ = 1872.3
costRec = 1872.3
```

4) See MATLAB code. In Figure 1, it can be observed that the Recursive approach is more robust to the introduced mismatch and the state variables remain closer to the equilibrium state $x_{eq} = [0, 0]^T$. Modifications to the variance changes the effect of the external disturbances and the capability of the control to make the state converge to the equilibrium.

Exercise 3 **Infinite Horizon cost-to-go**

1) See MATLAB code. The eigenvalues are

$$0.9900 + 0.0989i$$
$$0.9900 - 0.0989i$$

and the absolute value of both has to be smaller zero in order to be asymptotically stable. Here both are approximately 0.9949.

2) See MATLAB code, $J_0 = \sum_{k=0}^{N=100} x(k)^T Q x(k) \approx 96.3287$.

3) See MATLAB code, $\Psi(x(0)) = x(0)^T P x(0) \approx 149.4222$.

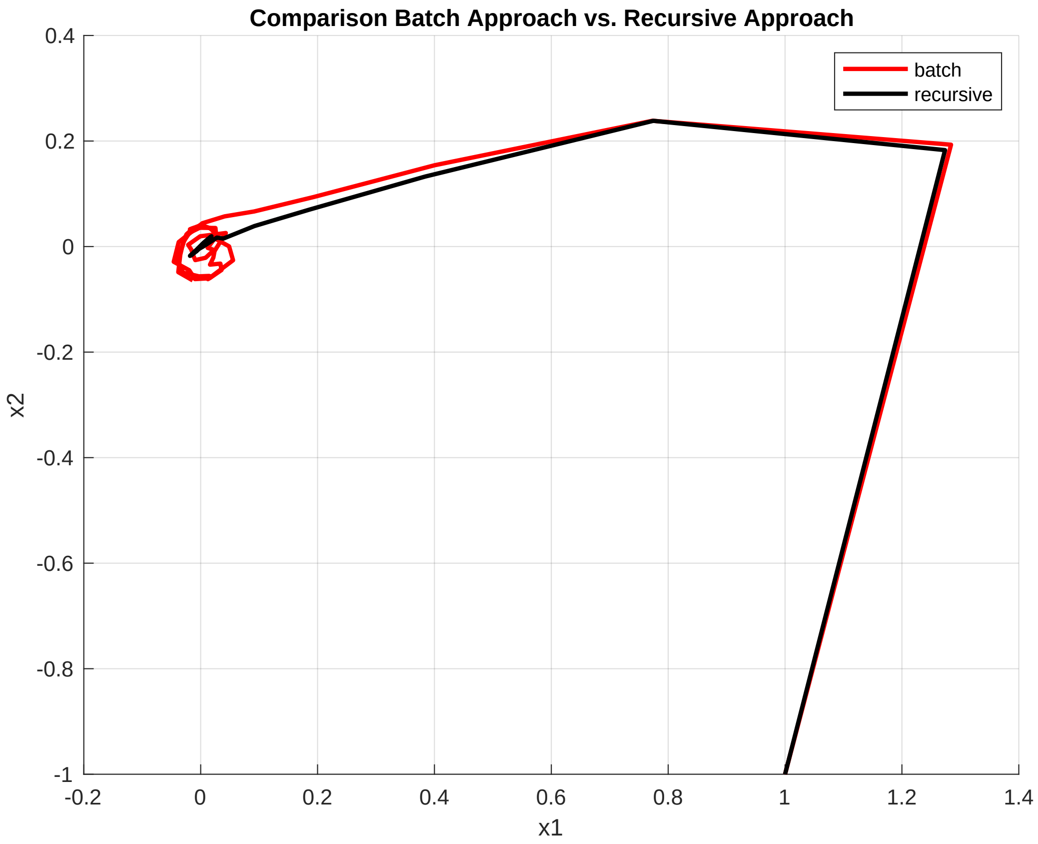

Figure 1: Comparison between Batch and Recursive Approach

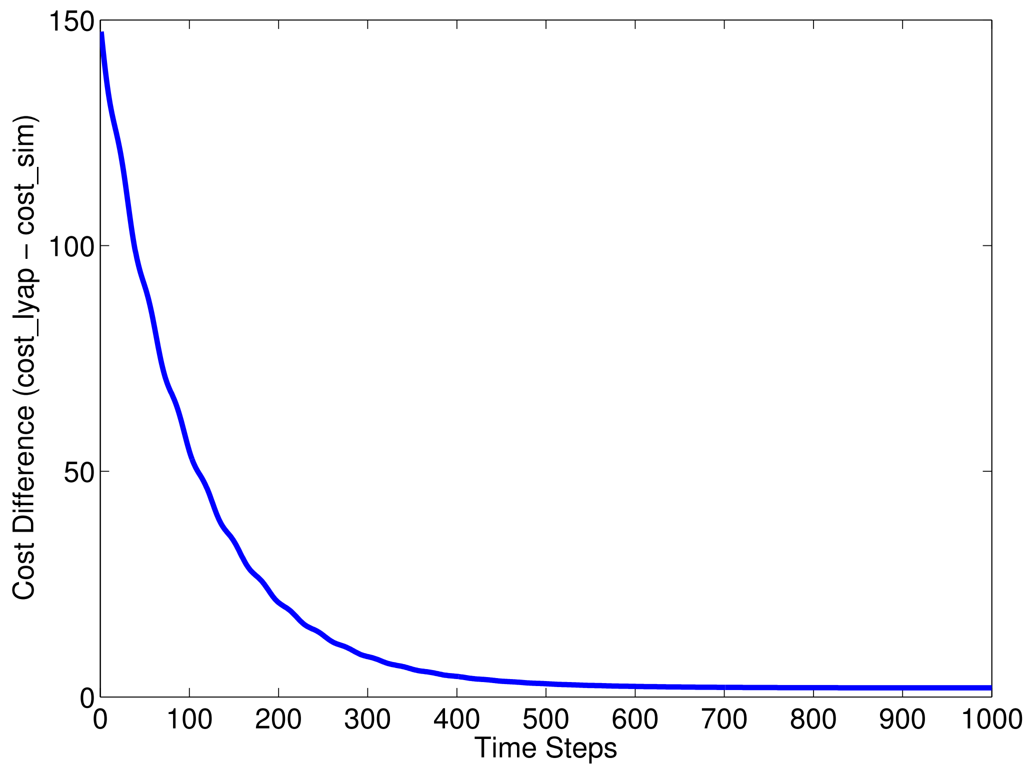

Figure 2: Cost function difference cost_lyap - cost_sim for increasing N